

# State of Developer Ecosystem 2024: Language

JetBrains survey of 23,262 developers • 2017–2024 usage, primary language, satisfaction, and don

**23,262**

Developers surveyed (global)

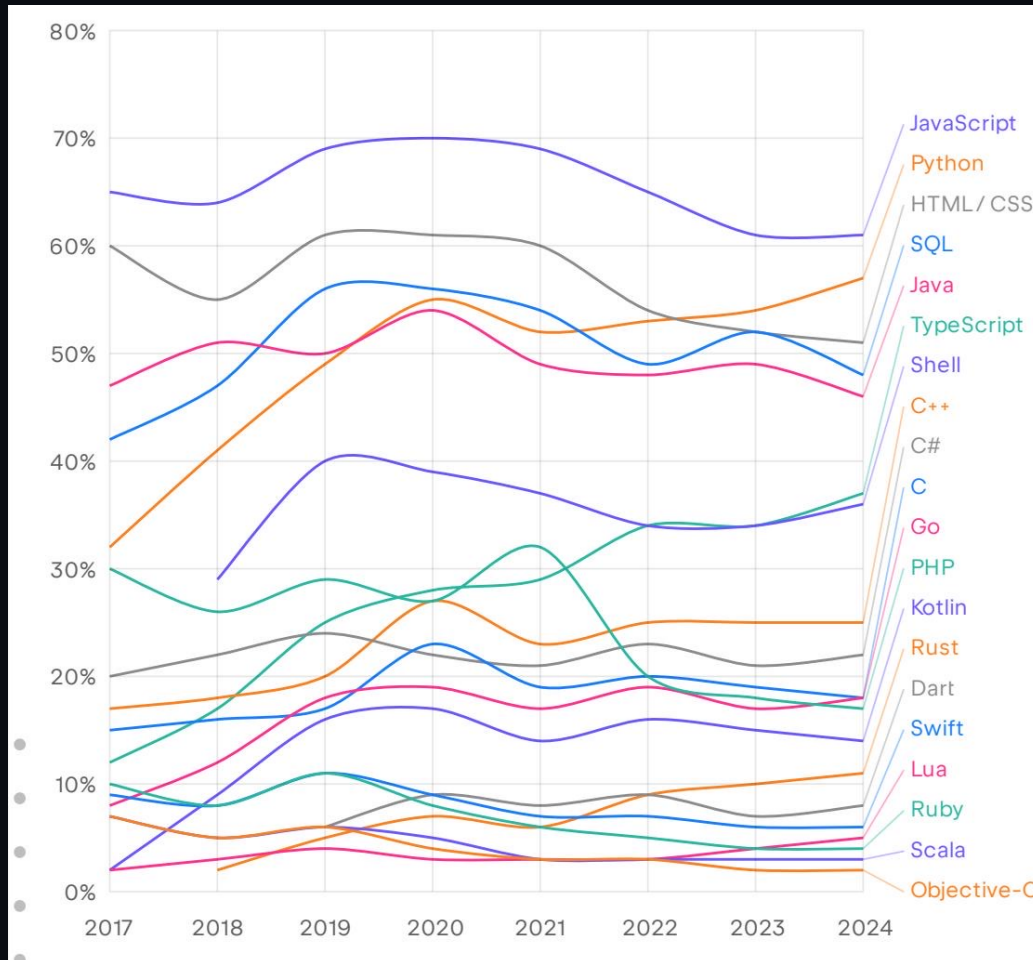
**2017–2024**

Eight years of language usage

**6 slides**

Executive-ready narrative

# Python Closes the Gap: 8 Years of Language



Usage = % of developers reporting they use the language

2017–2024 usage

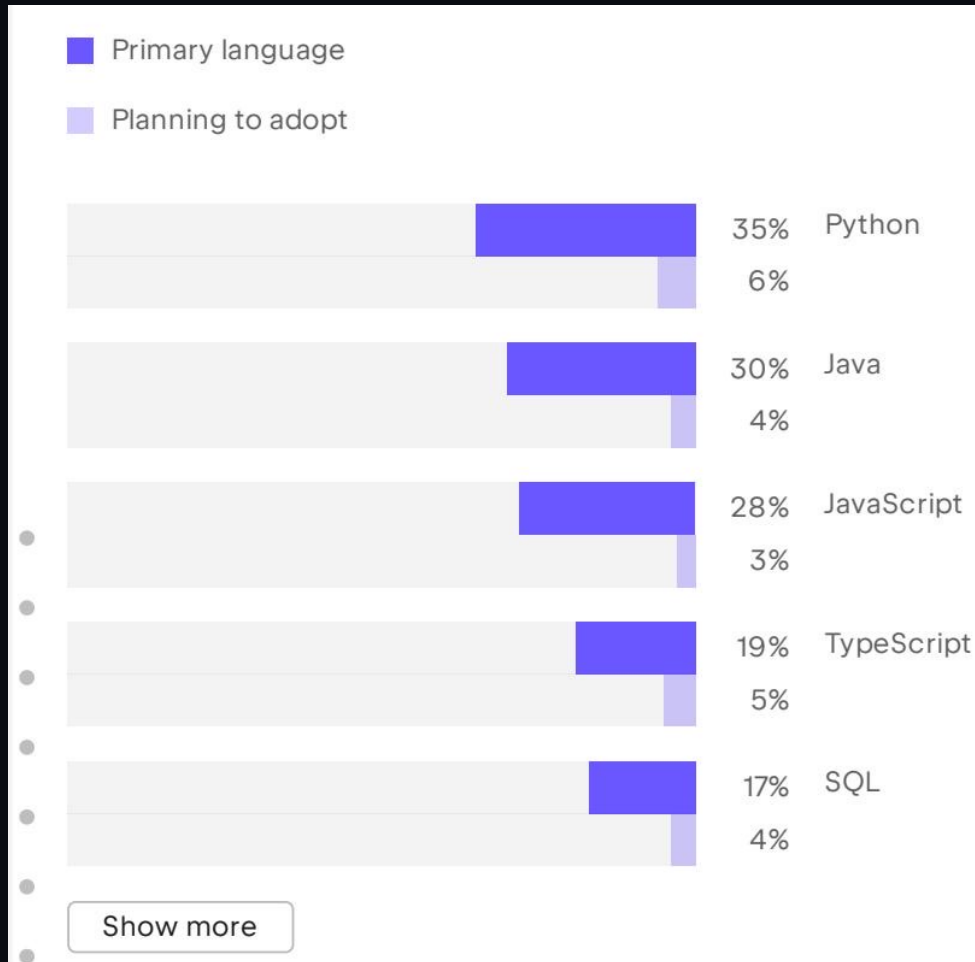
JavaScript: peaked at 70% (2020), down to 61% (2024) — 65% in 2017

Python: 32% (2017) → 57% (2024), narrowing the gap with JavaScript

TypeScript: 12% (2017) → 37% (2024), ~3× growth

PHP: 30% (2017) → 17% (2024), sustained decline

# Python Is Now the #1 Primary Language at



Adoption intent = % planning to adopt as a primary language

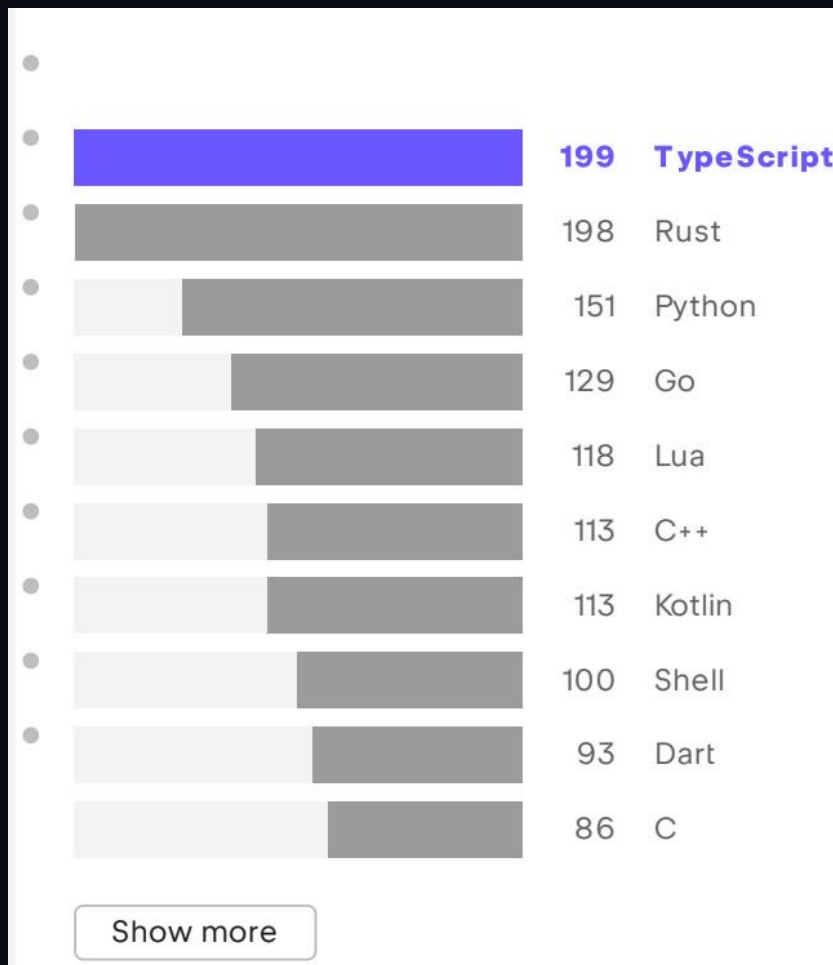
## Primary vs adopt

Primary language share:  
Python 35%, Java 30%,  
JavaScript 28%, TypeScript  
19%, SQL 17%

Adoption intent stays low  
( $\leq 6\%$ ): Python 6%, TypeScript  
5%, Java 4%, SQL 4%

Interpretation: ecosystem  
lock-in + expertise investment  
limits switching

# TypeScript and Rust Top Developer Satisfaction



## Top 10 (score)

1. TypeScript 199
2. Rust 198
3. Python 151
4. Go 129
5. Lua 118
6. C++ 113
7. Kotlin 113
8. Shell 100
9. Dart 93
10. C 86

Satisfaction scores from JetBrains survey (higher = more satisfied)

# Each Language Owns a Domain: AI Belongs

|   | Python | Java | JavaScript | TypeScript | SQL | C#  | C++ | PHP |                                                                |
|---|--------|------|------------|------------|-----|-----|-----|-----|----------------------------------------------------------------|
| • | 38%    | 48%  | 38%        | 47%        | 25% | 55% | 17% |     | Integrating with APIs and services                             |
| • | 35%    | 55%  | 38%        | 47%        | 37% | 63% | 34% |     | Application logic and workflows                                |
| • | 35%    | 17%  | 7%         | 6%         | 45% | 18% | 14% |     | Data processing and analytics                                  |
| • | 31%    | 5%   | 3%         | 4%         | 6%  | 6%  | 11% |     | Artificial intelligence and machine learning                   |
| • | 28%    | 49%  | 22%        | 29%        | 31% | 50% | 13% |     | Providing APIs and services                                    |
| • | 27%    | 5%   | 3%         | 2%         | 21% | 5%  | 6%  |     | Data science / Data analytics                                  |
| • | 26%    | 9%   | 10%        | 12%        | 5%  | 12% | 9%  |     | Integration with AI                                            |
| • | 22%    | 4%   | 9%         | 7%         | 11% | 8%  | 4%  |     | Ad hoc scripts                                                 |
| • | 19%    | 29%  | 23%        | 31%        | 11% | 36% | 30% |     | Libraries / Frameworks                                         |
| • | 16%    | 13%  | 9%         | 15%        | 4%  | 18% | 9%  |     | Automating software testing and quality assurance              |
| • | 14%    | 19%  | 58%        | 62%        | 8%  | 32% | 18% |     | User interface (handling user inputs and interactive elements) |
| • | 11%    | 10%  | 6%         | 6%         | 6%  | 15% | 19% |     | System tools and components                                    |
| • | 7%     | 9%   | 4%         | 4%         | 2%  | 12% | 18% |     | Network communication protocols                                |
| • | 6%     | 3%   | 4%         | 3%         | 1%  | 5%  | 13% |     | Audio or video processing                                      |
| • | 6%     | 9%   | 3%         | 4%         | 2%  | 9%  | 10% |     | Security, encryption, and cryptographic protocols              |
| • | 6%     | 1%   | 1%         | 0%         | 2%  | 2%  | 5%  |     | Conducting simulations and predictive modeling                 |
| • | 4%     | 4%   | 2%         | 1%         | 1%  | 7%  | 21% |     | Hardware interfacing and control systems                       |
| • | 2%     | 2%   | 4%         | 4%         | 0%  | 7%  | 15% |     | Rendering and processing graphics                              |

## Domain leaders

AI / ML: Python (31%)

Data processing & analytics: Python (17%)

Data science / analytics: Python (28%)

User interface:  
TypeScript (62%) /  
JavaScript (58%)

APIs & services: Java (49%)

System tools & components: C++ (27%)

Domain share = % selecting the language for that application area

# What This Means for Your Technology Strat

## 1) Double down on Python for AI/ML

Python leads AI/ML (31%) and data science (28%) and is rising in overall usage.

## 2) Standardize on TypeScript for UI

TypeScript dominates UI (62%) and ranks #1 in satisfaction (199).

## 3) Build infra options around Go/Rust

Go holds ~18% usage after rapid growth; Rust continues climbing to 11% and

## 4) Plan for legacy pool contraction

PHP usage declined from 30% (2017) to 17% (2024). Ruby fell from

## 5) Treat DX as a retention lever

Top satisfaction clusters around modern tooling and type safety (TypeScript, Rust).